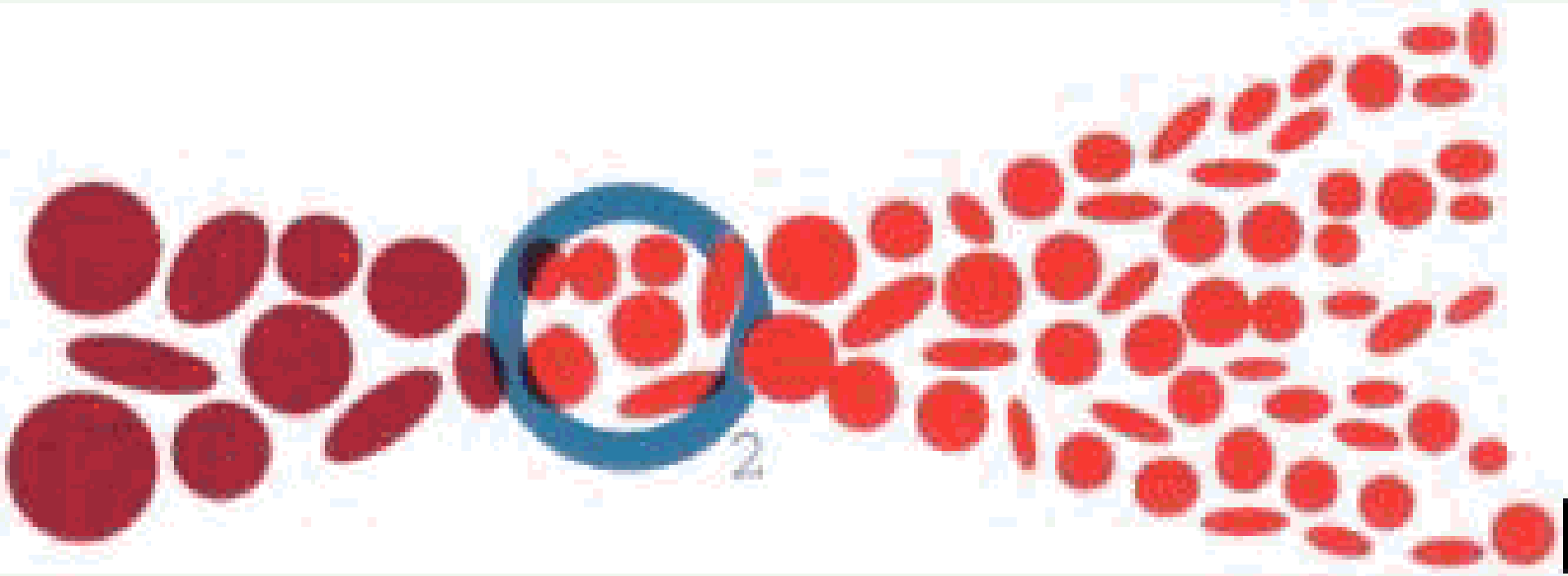
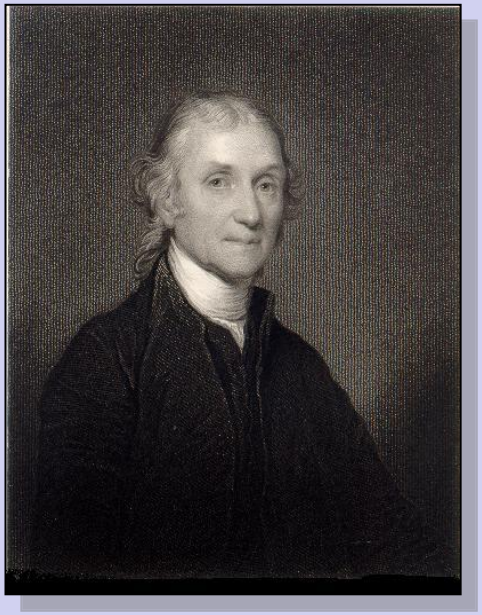
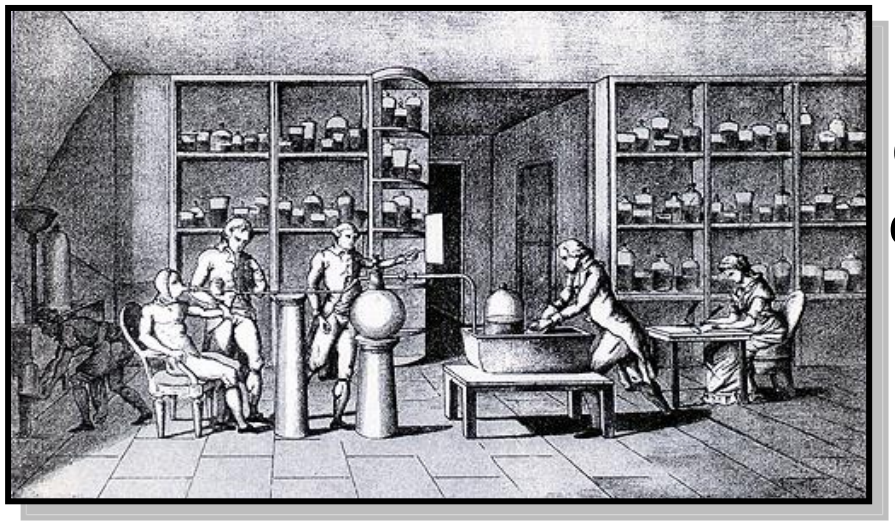
OXIVENACIÓN



Inyección lenta y pausada de oxígeno puro medicinal en una vena, con fines terapéuticos

1774: Priestley descubre el oxígeno y describe su función en la combustión y en la respiración de los seres vivos





1789: Antoine Laurent Lavoisier describe la participación del oxígeno en los procesos de combustión, en la respiración y en la oxidación de metales. Reporta efectos tóxicos del oxígeno a altas concentraciones

Journal of Clinical Anesthesia (2006) 18, 388–395



ELSEVIER

Journal of Clinical Anesthesia

Special article

The history of subcutaneous oxygen therapy☆

Timothy B. Curry MD, PhD*, Douglas R. Bacon MD, MA, Richard H. Rho MD

[7]. Initially, the best method of oxygen administration was unknown; the effective use of inhaled oxygen would not be understood or even possible for many years. Consequently, many means of supplying gas to the body were tried, including instilling it orally and rectally and injecting it intravenously, intraperitoneally, and subcutaneously [8].

Spallanzani (1729 – 1799): Completa absorción de oxígeno inyectado bajo la piel de animales



Beddoes (1799): En el Instituto Neumático de Bristol, utilizó oxígeno subcutáneo en seres humanos



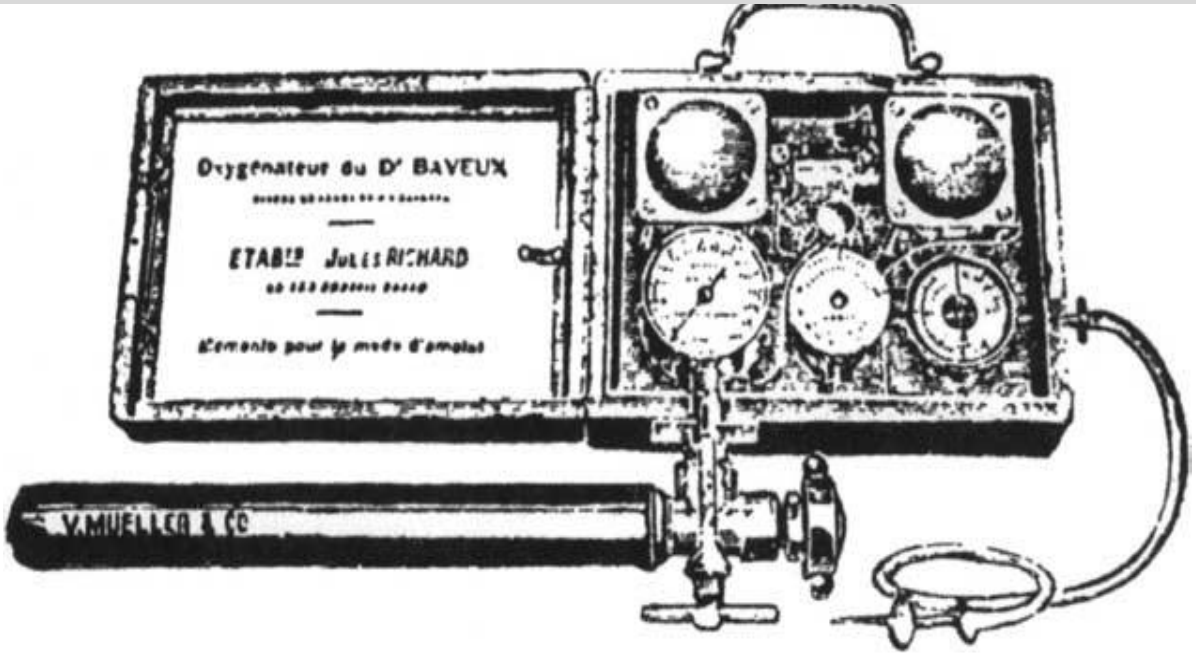
Breath of life: the evolution of oxygen therapy

C Grainge MRCP

J R Soc Med 2004;**97**:489–493

This first description of a step change in the clinical delivery of oxygen was published in competition with many papers from less perceptive individuals advocating methods for administering the gas that have not stood the test of time. Contemporary publications advocated oxygen administration via the stomach for resuscitation, via the urethra for inflammatory conditions and by enema for the treatment of gallstones (a proposal published in the *Journal of the American Medical Association* by JH Kellogg MD).[8]

1910: Bayeux describe efectos del uso de oxígeno subcutáneo en hipoxia por altitud. Diseña y comercializa dispositivo portátil.



Peróxido de sodio en contacto con agua

Howitt y Derose (1912 – 1914): Oxígeno subcutáneo en casos de TBC, uremia, eclampsia, diabetes, enfisema, tosferina

THE SUBCUTANEOUS INJECTION OF OXYGEN GAS, by H.O. Howitt, M.D., L.R.C.P. (Lon.), M.R.C.S. (Eng.), Guelph. The Canadian Medical Association Journal, 1914, Volume 4; pages 983-985

The following is a description of the apparatus and the method of its use: Oxygen is generated from sodium peroxide coming in contact with water in a closed cylinder, and the gas set free escapes by means of a long rubber tube, at the distal end of which is an aspirating needle which is inserted into the subcutaneous tissue. The tube is then oiled and by a pumping action of the hand, sufficient force is used to raise a lump once or twice as large as the closed fist.

The oxygen injected by this method forms a local emphysema which lasts for hours and may travel over the surface of the body. Apparently the body only absorbs that which is necessary. It is really the formation of an artificial lung. I have never known a single ill effect to follow its use; in every case where the breathing was laboured, the patient experienced some relief. In some cases, the ones I have mentioned, the results were almost as wonderful as a chapter from the "Arabian Nights," and were witnessed by

1902: Inyección de oxígeno intravenoso en perros (150 – 300 mL)

1916: Tunniclife reporta uso de oxígeno disuelto en solución salina, en aplicación endovenosa.

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The Lancet, Volume 188, Issue 4851, Pages 321 - 323, 19 August 1916
doi:10.1016/S0140-6736(00)97237-4 Cite or Link Using DOI

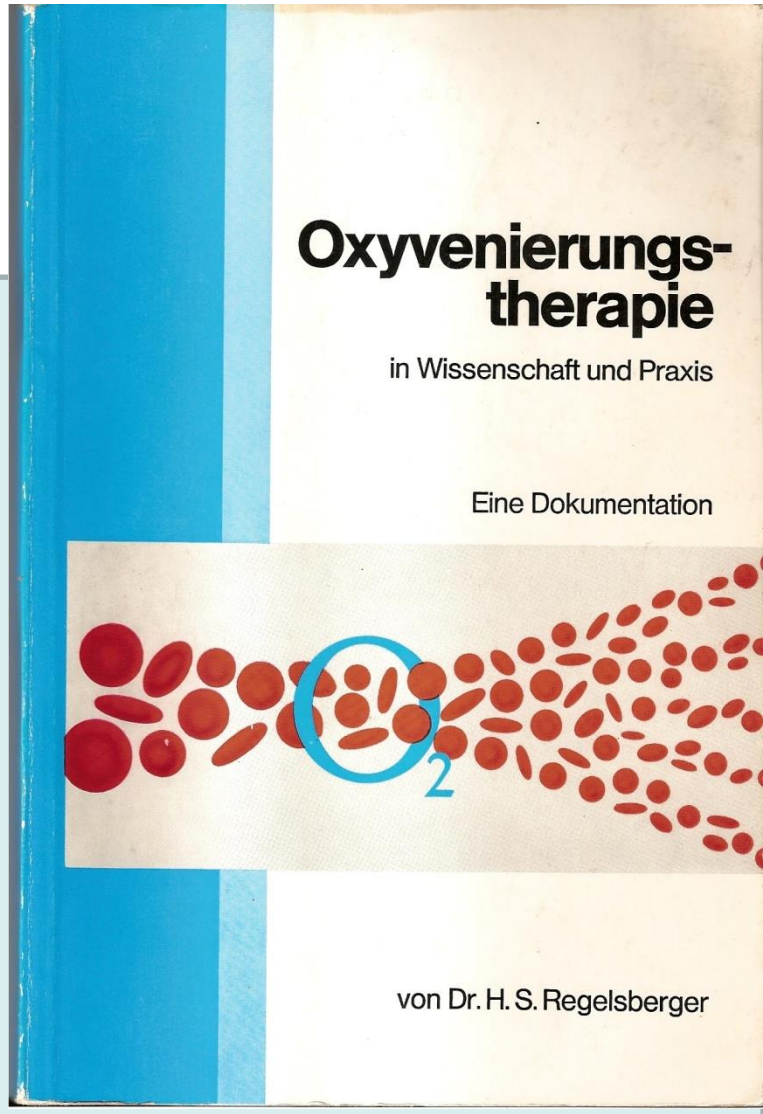
< Previous Article | Next Article >

THE INTRAVENOUS INJECTION OF OXYGEN GAS AS A THERAPEUTIC MEASURE.

F.W. Tunnicliffe M.D. LOND. , G.F. Stebbing M.B., B.S. LOND.

The object of this preliminary note is not to indicate the specific pathological conditions in which the intravenous injection of oxygen should be used, but simply to point out that in those cases in which the inhalation of oxygen either cannot be practiced or gives inadequate relief, the intravenous method of oxygen administration is available to the clinician and will give therapeutic results.

- Dr. Helmut Regelsberger. Neurocirujano. 1945. Universitat Köln.
- Inyección endovenosa de oxígeno en paciente con tumor cerebral.
- Creación del Instituto de investigación. 1953
- Publicación del libro Terapia de Oxivenación ante la ciencia y en la práctica
- Sociedad alemana para la terapia de oxivenación. 1959

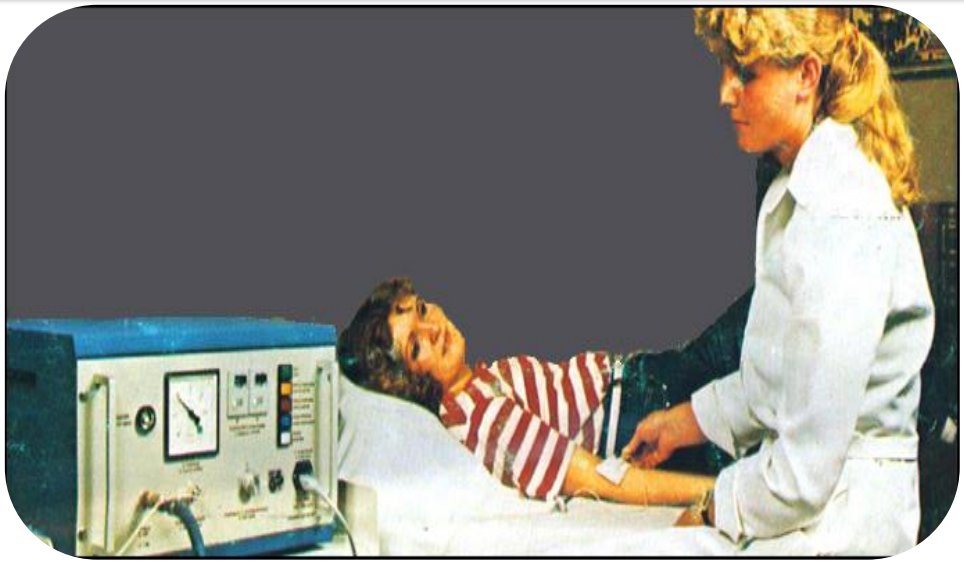




- 1985: Dr. Jorge Piñeros Corpas. Uso de oxígeno disuelto en solución para aplicación endovenosa.

Métodos de oxivenación

1. Método Regelsberger

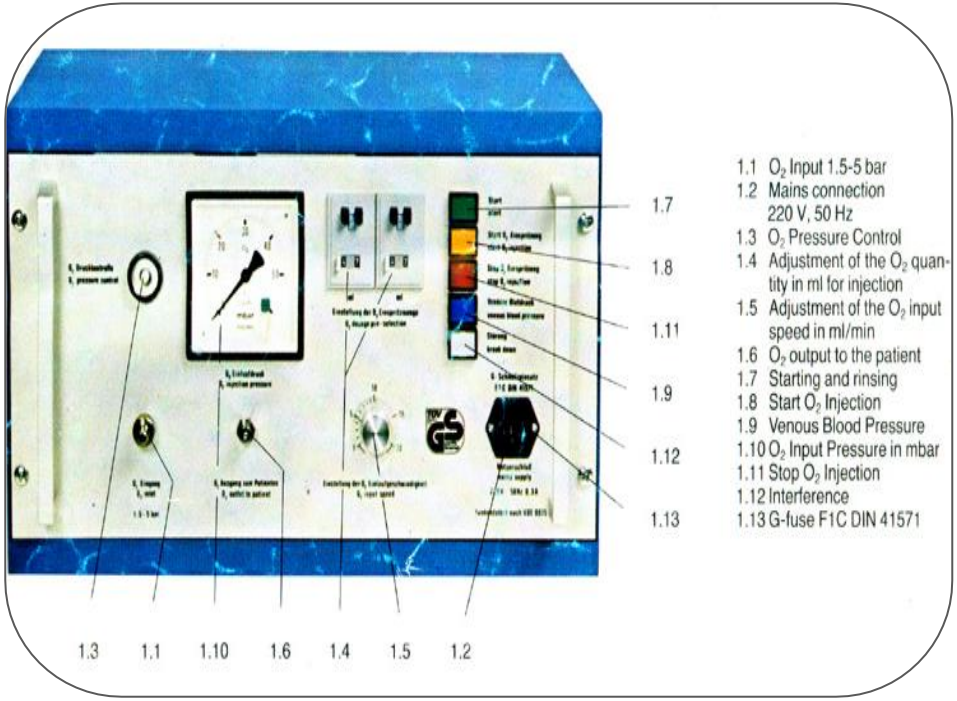




- Equipo de oxígeno peristático
- Inyección de oxígeno a velocidad entre 0.5 y 4 mL/min.
- Velocidad sugerida: 2 mL/min

https://www.youtube.com/watch?v=DnMv_CqFTyI

https://www.gesundheitspraxis-menden.de/oxyvenierung.htm



1.1 O_2 Input 1.5-5 bar
1.2 Mains connection 220 V, 50 Hz
1.3 O_2 Pressure Control
1.4 Adjustment of the O_2 quantity in ml for injection
1.5 Adjustment of the O_2 input speed in ml/min
1.6 O_2 output to the patient
1.7 Starting and rinsing
1.8 Start O_2 Injection
1.9 Venous Blood Pressure
1.10 O_2 Input Pressure in mbar
1.11 Stop O_2 Injection
1.12 Interference
1.13 G-fuse F1C DIN 41571



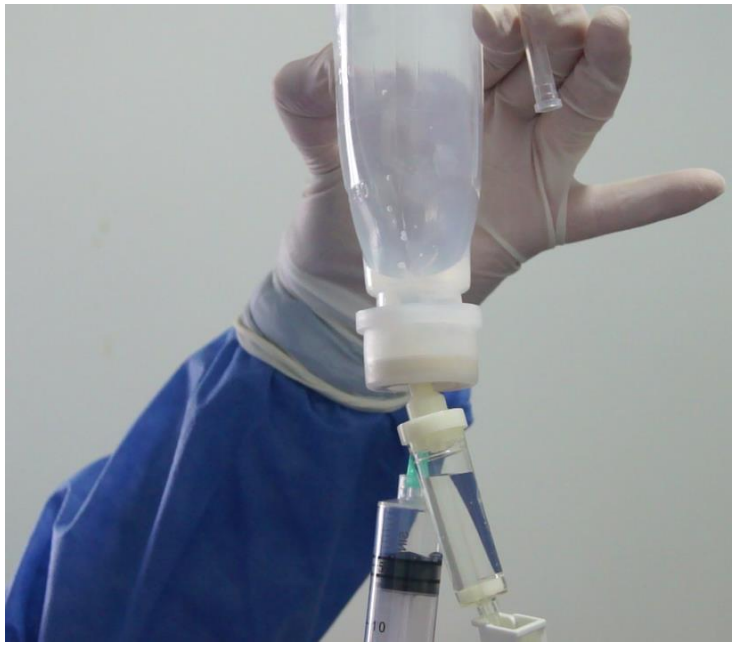


https://www.youtube.com/watch?v=dsYq4X7vs-I

2. Método Corpas

Oxígeno disuelto en solución para aplicación endovenosa





Técnica.
Método Corpas

Materiales

1. Oxígeno medicinal puro
2. Dextrosa en Agua Destilada o SSN 0,9% 250 mL.
3. Equipo de venoclisis
4. Yelco







Dosificación método Corpas

- Sesión 1: 5 mL
- Sesión 2: 10 mL
- Sesión 3: 15 mL
- Aumentar el volumen 5 mL en cada sesión hasta Volumen máximo de 50 mL.
- Pasar por vena periférica a 60 gotas por minuto
- Frecuencia: 1 a 2 sesiones semanales (10 sesiones)

Procedimiento



Soluciones Intravenosas Hiperoxigenadas (HOS).
Evidencia científica.

Sun YH: Opening up new channels for delivery of oxygen to prevent and reduce the hypoxic injury. Natl Med J China 2004; 84: 79 – 82 [in Chinese].

International Journal of Nanomedicine

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Open Access Full Text Article

ORIGINAL RESEARCH

Oxygen supersaturated fluid using fine micro/nanobubbles

This article was published in the following Dove Press journal:
International Journal of Nanomedicine
23 September 2014
Number of times this article has been viewed

Noriaki Matsuki[1]
Takuji Ishikawa[2]
Shingo Ichiba[3]
Naoki Shiba[3]
Yoshihito Ujike[3]
Takami Yamaguchi[4]

[1]Department of Biomedical Engineering, Graduate School of Engineering, Okayama University of Science, Okayama, [2]Department of Bioengineering and Robotics, Graduate School of Engineering, Tohoku University, Sendai, [3]Department of Emergency and Critical Care Medicine, Okayama University Hospital, Okayama, [4]Graduate School of Biomedical Engineering, Tohoku University, Sendai, Japan

Abstract: Microbubbles show peculiar properties, such as shrinking collapse, long lifetime, high gas solubility, negative electric charge, and free radical production. Fluids supersaturated with various gases can be easily generated using microbubbles. Oxygen microbubble fluid can be very useful for oxygen delivery to hypoxic tissues. However, there have been no reports of comparative investigations into adding fluids containing oxygen fine micro/nanobubbles (OFMNBs) to common infusion solutions in daily medical care. In this study, it was demonstrated that OFMNBs can generate oxygen-supersaturated fluids, and they may be sufficiently small to infuse safely into blood vessels. It was found that normal saline solution is preferable for generating an oxygen-rich infusion fluid, which is best administered as a 30-minute intravenous infusion. It was also concluded that dextran solution is suitable for drug delivery substances packing oxygen gas over a 1-hour intravenous infusion. In addition, normal saline solution containing OFMNBs was effective for improving blood oxygenation. Thus, the use of OFMNB-containing fluids is a potentially effective novel method for improving blood oxygenation in cases involving hypoxia, ischemic diseases, infection control, and anticancer chemoradiation therapies.

Keywords: microbubble, fine micro/nanobubble, nanobubble, oxygenation, fluid oxygenation

Format: Abstract ▾ Send to ▾

Molecules. 2018 Aug 31;23(9). pii: E2210. doi: 10.3390/molecules23092210.

Oxygen-Carrying Micro/Nanobubbles: Composition, Synthesis Techniques and Potential Prospects in Photo-Triggered Theranostics.

Khan MS[1], Hwang J[2], Lee K[3], Choi Y[4], Kim K[5], Koo HJ[6], Hong JW[7,8], Choi J[9].

⊕ Author information

Abstract

Microbubbles and nanobubbles (MNBs) can be prepared using various shells, such as phospholipids, polymers, proteins, and surfactants. MNBs contain gas cores due to which they are echogenic and can be used as contrast agents for ultrasonic and photoacoustic imaging. These bubbles can be engineered in various sizes as vehicles for gas and drug delivery applications with novel properties and flexible structures. Hypoxic areas in tumors develop owing to an imbalance of oxygen supply and demand. In tumors, hypoxic regions have shown more resistance to chemotherapy, radiotherapy, and photodynamic therapies. The efficacy of photodynamic therapy depends on the effective accumulation of photosensitizer drug in tumors and the availability of oxygen in the tumor to generate reactive oxygen species. MNBs have been shown to reverse hypoxic conditions, degradation of hypoxia inducible factor 1α protein, and increase tissue oxygen levels. This review summarizes the synthesis methods and shell compositions of micro/nanobubbles and methods deployed for oxygen delivery. Methods of functionalization of MNBs, their ability to deliver oxygen and drugs, incorporation of photosensitizers and potential application of photo-triggered theranostics, have also been discussed.

KEYWORDS: microbubbles; nanobubbles; oxygen delivery; photoacoustic imaging; reactive oxygen species (ROS); ultrasonic imaging

PMID: 30200336 PMCID: PMC6225314 DOI: 10.3390/molecules23092210

Check for updates

The Journal of International Medical Research
2011; 39: 1589 – 1606

Hyperoxygenated Solutions in Clinical Practice: Preventing or Reducing Hypoxic Injury

H Xu[a], Y Li[a], W Gao, H Zhang and L-X Xu

Department of Anaesthesiology, School of Stomatology, The Fourth Military Medical University, Xi'an, China

In cases of hypoxia, oxygen can be supplied via a number of methods including face masks, nasal cannulae, hyperoxygenated oxygen chambers and mechanical ventilation. Administering oxygen via the respiratory tract is, however, limited in respiratory diseases such as pulmonary fibrosis, pneumoconiosis and severe acute respiratory syndrome, or following the inhalation of asphyxiating poisons. This has led to research into new methods of supplying oxygen that bypass the lungs. Research has investigated the efficacy of intravenous hyperoxygenated solutions (HOS) as auxiliary oxygen supplies in several hypoxic states including cardiopulmonary resuscitation, respiratory failure, cerebrovascular disease, myocardial ischaemia, severe acute respiratory syndrome, poisoning, neonatal hypoxia, altitude sickness, large burns and traumatic haemorrhagic shock. Much of the research has taken place in China and more than 3.5 million hypoxic patients have received HOS, with good therapeutic effects. This review summarizes the literature supporting the use of this novel treatment.

KEY WORDS: Hyperoxygenated solution; Oxygen therapy; Hypoxia



FIGURE 1: The GY-1 dissolved oxygen apparatus used to produce hyperoxygenated solutions for medical infusions

The mean ± SD PaO2 in water is 20.5 ± 0.4 kPa (153.8 ± 3.0 mmHg) before oxygen is dissolved.

After HOS is prepared, there is a significant increase in the mean ± SD PaO2 to 106.3 ± 7.8 kPa (797.3 ± 58.5 mmHg).

The PaO2 in HOS is, therefore, 5.2 times that of conventional liquid, 8.3 times that of arterial blood and 21.3 times that of venous blood.

Hyperoxygenated Solutions in Clinical Practice: Preventing or Reducing Hypoxic Injury (sagepub.com)

Modelos animales

ELSEVIER

Journal of Surgical Research
Volume 164, Issue 1, November 2010, Pages 131-138



Shock/Sepsis/Trauma/Critical Care

Intravenous Administration of Hyperoxygenated Solution Attenuates Pulmonary Edema Formation in Phosgene-Induced Acute Lung Injury in Rabbits

Ling Wang M.D. *, 2 ✉, Chunran Liu M.D. *, 2, Hui Zhang M.D. *, Changjun Gao M.D. †, Wei Chai M.D. †, Ruifen Xu M.D. *, Hui-xia Wang M.D. ‡, Lixian Xu M.D. * ✉

Intravenous HOS infusion after phosgene exposure can clearly lessen phosgene-induced lung edema formation, lipid peroxidatic reaction, and ameliorate hypoxemia associated with phosgenismus; it is a safe, simple, and effective measure to protect animals from phosgene-induced lung injury.

L. Wang, C. Liu, H. Zhang, C. Gao, W. Chai, R. Xu, H.X. Wang, L. Xu
Intravenous administration of hyperoxygenated solution attenuates pulmonary edema formation in phosgene-induced acute lung injury in rabbits. J. Surg. Res., 164 (2010), pp. 131-138

Journal of Plastic, Reconstructive & Aesthetic Surgery (2020) 73, 975–982



JPRAS
An International Journal of Surgical Reconstruction
www.JPRASurg.com

Hyperoxygenated solution improves tissue viability in an avulsion injury flap model



Jianzhang Wang[1], Zhangqiang Tuo[1], Juan Zhang, Peng Guo, Baoqiang Song*

Department of Plastic and Reconstructive Surgery, Xijing Hospital, Fourth Military Medical University, No.127 Changle West Road, Xi'an, Shanxi Province 710032, People's Republic of China

Received 25 June 2019; accepted 24 November 2019

We succeeded in developing an avulsion injury model and demonstrated that HOS could improve the survival of the avulsion injury flaps in rats by effectively increasing the local oxygen content and blood perfusion and ameliorating inflammatory damage

Jianzhang Wang, Zhangqiang Tuo, Juan Zhang, Peng Guo, Baoqiang Song,
Hyperoxygenated solution improves tissue viability in an avulsion injury flap model,
Journal of Plastic, Reconstructive & Aesthetic Surgery,
Volume 73, Issue 5,
2020,
Pages 975-982,
ISSN 1748-6815,
https://doi.org/10.1016/j.bjps.2019.11.029.
(https://www.sciencedirect.com/science/article/pii/S1748681519305170)

Circulation
Volume 96, Issue 12, 16 December 1997; Pages 4385-4391
https://doi.org/10.1161/01.CIR.96.12.4385

ARTICLE

Aqueous Oxygen

A Highly O_2-Supersaturated Infusate for Regional Correction of Hypoxemia and Production of Hyperoxemia

J. Richard Spears, Bing Wang, Xiaojun Wu, Petar Prcevski, Alice J. Jiang, Ali D. Spanta, Richard J. Crilly, and Giles J. Brereton

The results of our studies support the hypotheses that correction of arterial hypoxemia and production of hyperoxemia on a regional basis are achievable with an AO infusion. Specific hypotheses regarding the effects of hyperoxemia on ischemic tissue function and metabolism can now be addressed with this catheter-based approach without concern for pulmonary oxygen toxicity. AO infusion on room air breathing are similar to those noted with oxygen breathing without the infusion. The mean arterial and venous oxygen tensions achieved during AO infusion on oxygen breathing approach values anticipated under hyperbaric conditions.

Aqueous Oxygen | Circulation (ahajournals.org)

Data in Brief

journal homepage: www.elsevier.com/locate/dib

Data Article

Data on a simple method for producing a solution that contains a high partial pressure of oxygen and a low partial pressure of carbon dioxide

Yoshihiro Tange *, Shigenori Yoshitake

Department of Medical Engineering, Kyushu University of Health and Welfare, Japan

ARTICLE INFO

Article history:
Received 2 February 2018
Accepted 27 February 2018
Available online 8 March 2018

ABSTRACT

The data presented here shows a simple method for producing a solution that contains a high partial pressure of oxygen (pO_2) and a low partial pressure of carbon dioxide (pCO_2). This novel solution was created by simply injecting oxygen gas into conventional supplemental bicarbonate fluid for renal replacement therapy. We compared the gas profiles of the novel solution and the conventional fluid in vitro. There was a significant increase in pO_2 and pH, and a significant decrease in pCO_2 in the experimental solution, in each of which an additional volume of oxygen was injected. The method shown here is capable of facilitating an increase of pO_2 and decrease of pCO_2 by using a closed fluid bag without any special devices.

Data on a simple method for producing a solution that contains a high partial pressure of oxygen and a low partial pressure of carbon dioxide - ScienceDirect

Journal of Artificial Organs (2018) 21:392–395
https://doi.org/10.1007/s10047-018-1022-9

TECHNICAL FORUM

Others

Simple method to make a supersaturated oxygen fluid

Yoshihiro Tange[1] · Shigenori Yoshitake[1] · Shingo Takesawa[1]

Received: 7 September 2017 / Accepted: 19 January 2018 / Published online: 22 January 2018
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Abstract

Intravenous oxygenation has demonstrated significant increase in partial pressure of oxygen (PO_2) in animal models. A highly dissolved oxygen solution might be able to provide a sufficient level of oxygen delivery to the tissues and organs in patients with hypoxia. However, conventional fluid oxygenation methods have required the use of original devices. If simpler oxygenation of a solution is possible, it will be a useful strategy for application in clinical practice. We simply developed its administration by injection of either air or oxygen gas into conventional saline. We determined the PO_2 values in the solutions in comparison with conventional saline in vitro. To examine the effects of the administration of the new solutions on the blood gas profile, we diluted bovine blood with either conventional or the new solutions and analyzed PO_2, oxygen saturation (SO_2) and total oxygen content. PO_2 levels in the blood and new solution mixture significantly increased with each additional injected gas volume. Significant increases in the PO_2 and SO_2 of the bovine blood were found in those blood samples with the new solution, as compared with those with the control solution. These results suggest that this solution promotes oxygen delivery to the hypoxic tissue and recovery from hypoxia. This method is simpler and easier than previous methods.

Keywords Supersaturated oxygen solution · Blood oxygenation · Hypoxia

Original Article | Published: 02 January 2019

Oxygen delivery to the blood stream by the dialysis fluid during continuous renal replacement therapy ex vivo

Yoshihiro Tange & Shigenori Yoshitake

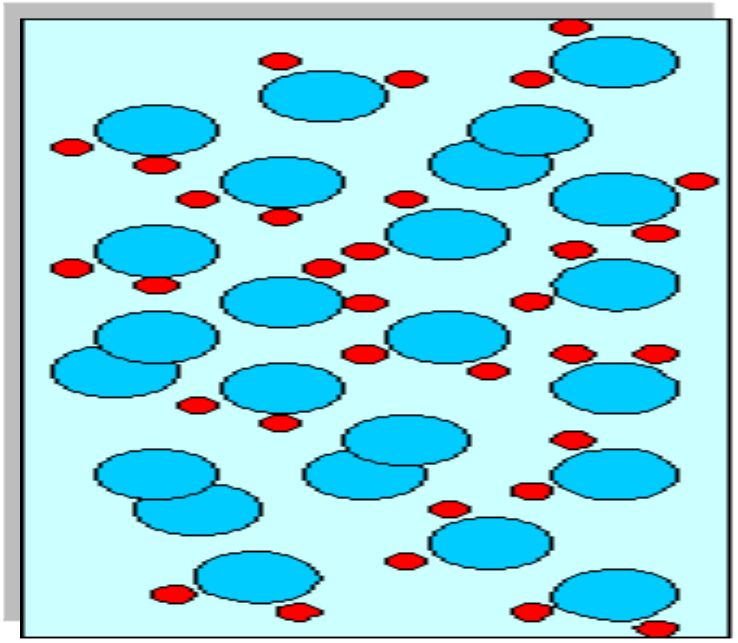
Journal of Artificial Organs **22**, 104–109 (2019) | Cite this article

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We prepared a saturated oxygen solution that enables a significant increase in the partial pressure of oxygen and oxygen delivery, and a significant decrease in the partial pressure of carbon dioxide in contrast to that achieved from using the conventional solution for continuous hemodiafiltration. Compared to the conventional solution, this solution can deliver a significant amount of oxygen from the dialysis fluid to the blood stream.

Tange, Y., Yoshitake, S. Oxygen delivery to the blood stream by the dialysis fluid during continuous renal replacement therapy ex vivo. *J Artif Organs* **22**, 104–109 (2019). https://doi.org/10.1007/s10047-018-1085-7

Los gases se solubilizan en los líquidos de acuerdo con: presión, temperatura, coeficiente de solubilidad y tipo de solvente.



Coeficientes de solubilidad en agua a 1 atm y 37 °C

Oxígeno……0,024
CO_2……0,570
Nitrógeno…….. 0,012
Helio…….. 0,008
Monóxido de Carbono…….0,018



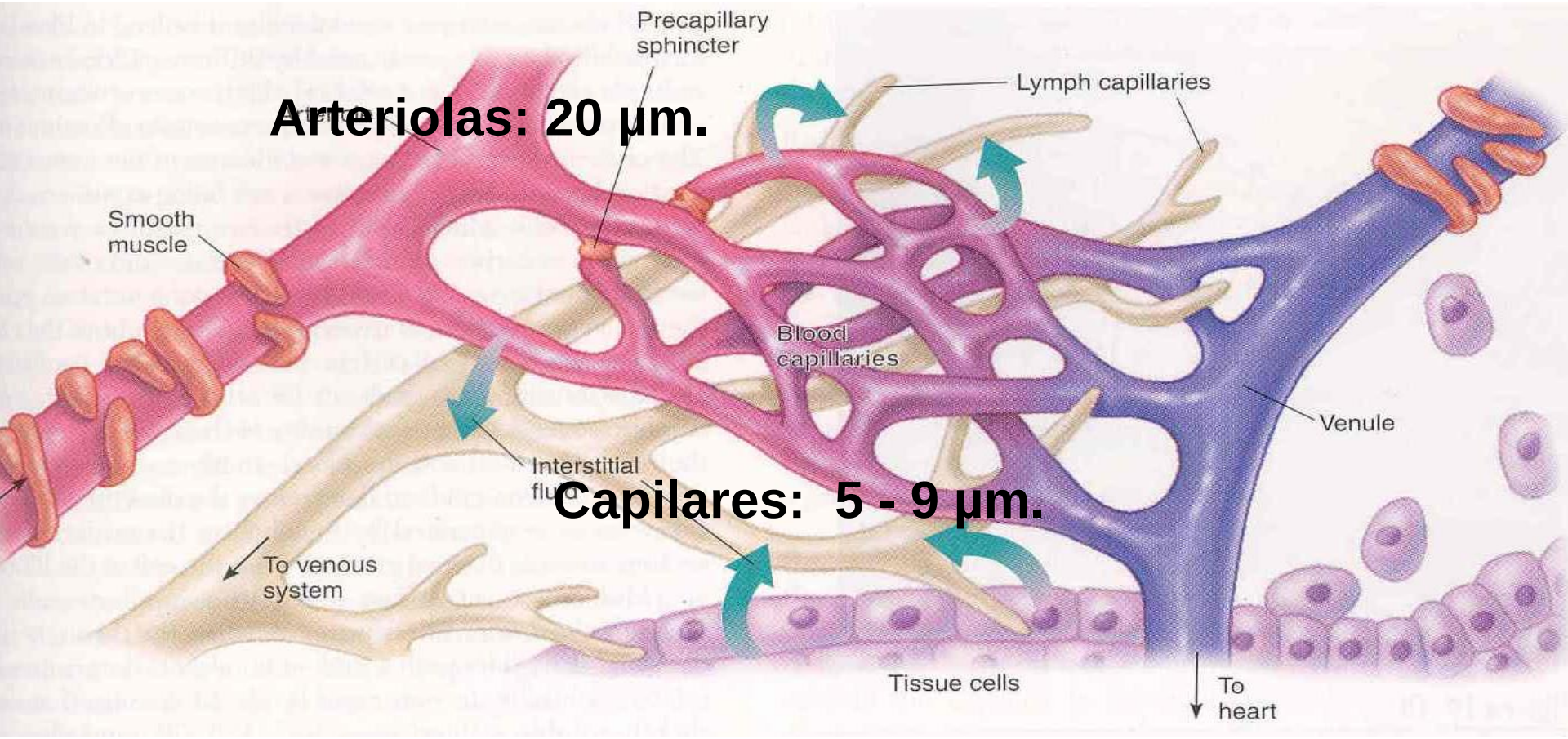
- La solubilidad de los gases en los líquidos disminuye al aumentar la temperatura.
- La solubilidad de los gases en los líquidos disminuye al aumentar las sales.

TABLE 1:
Comparison of the densities of oxygen, ozone and air and their solubilities in water across a range of temperatures

Gas	Density (g/l)	Solubility (ml/l) at various temperatures			
		0 °C	10 °C	20 °C	30 °C
Oxygen	1.492	49.3	38.4	31.4	26.7
Ozone	2.143	641.0	520.0	368.0	233.0
Air	1.293	28.8	23.6	18.7	16.1

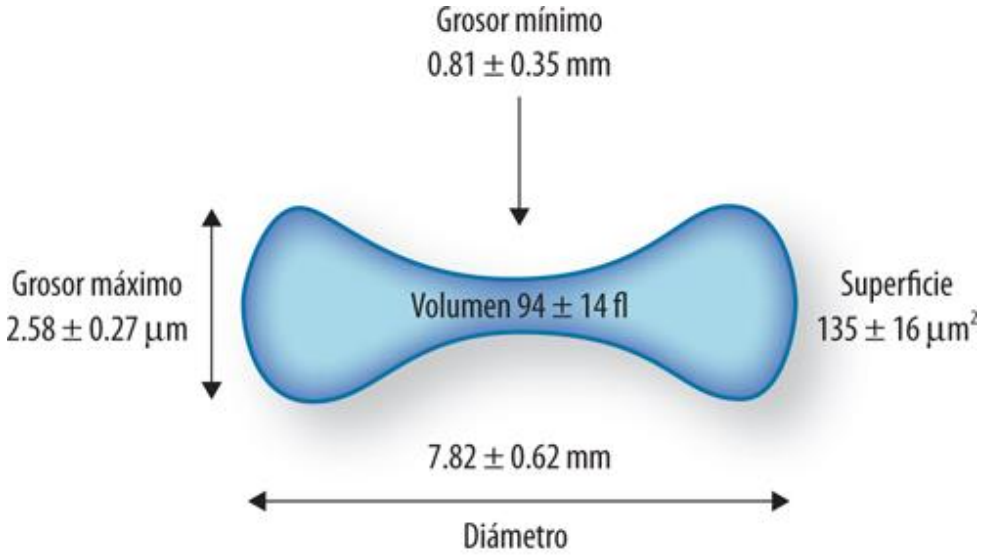
Hyperoxygenated Solutions in Clinical Practice: Preventing or Reducing Hypoxic Injury (sagepub.com)

Efectos biológicos de la oxivenación

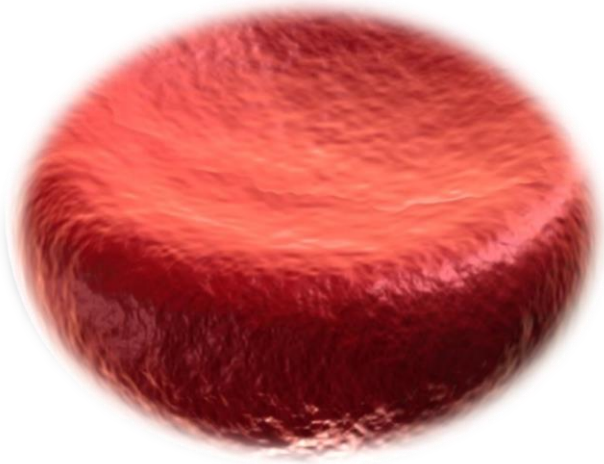


Superficie de intercambio: 500 and 700 m2

1. Efectos sobre el Eritrocito



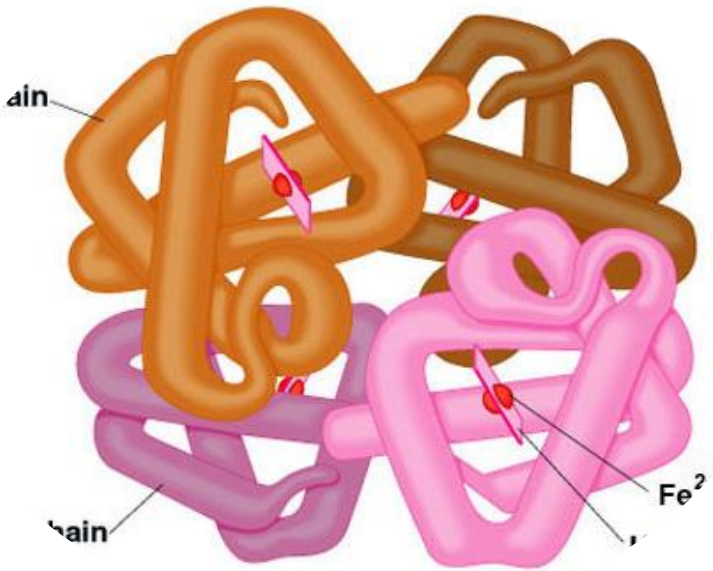
Fuente: Jesús A. Fernández-Tresguerres: *Fisiología humana*, 4e:
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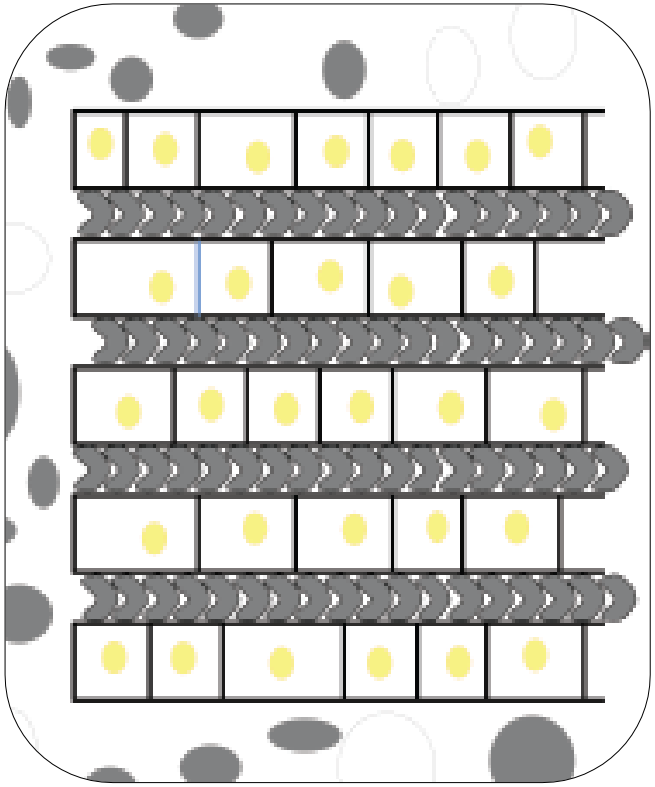


Diámetro: 7.8 µ
Espesor periferia: 2.5 µ
Espesor centro: 0.8 µ
Volumen medio: 90 µ3

https://accessmedicina.mhmedical.com/content.aspx?bookid=1858&sectionid=134364550

- 30 billones de eritrocitos
- 1 eritrocito: 265 millones de moléculas de hemoglobina
- 1 hemoglobina: 4 moléculas de oxígeno
- 1 g de Hb se combina con 1.34 mL de oxígeno
- 15 g de Hb se combinan con 20,1 mL de oxígeno







Microcirculación

- 30 billones de eritrocitos
- Diámetro capilar interno: 7.5 a 12 µ.
- Red capilar: 100.000 Km de longitud y 6300 m^2 de superficie
- “El eritrocito se deforma de cualquier manera. Le sobra membrana celular”

  Guyton
- La deformabilidad es una función dependiente de ATP.



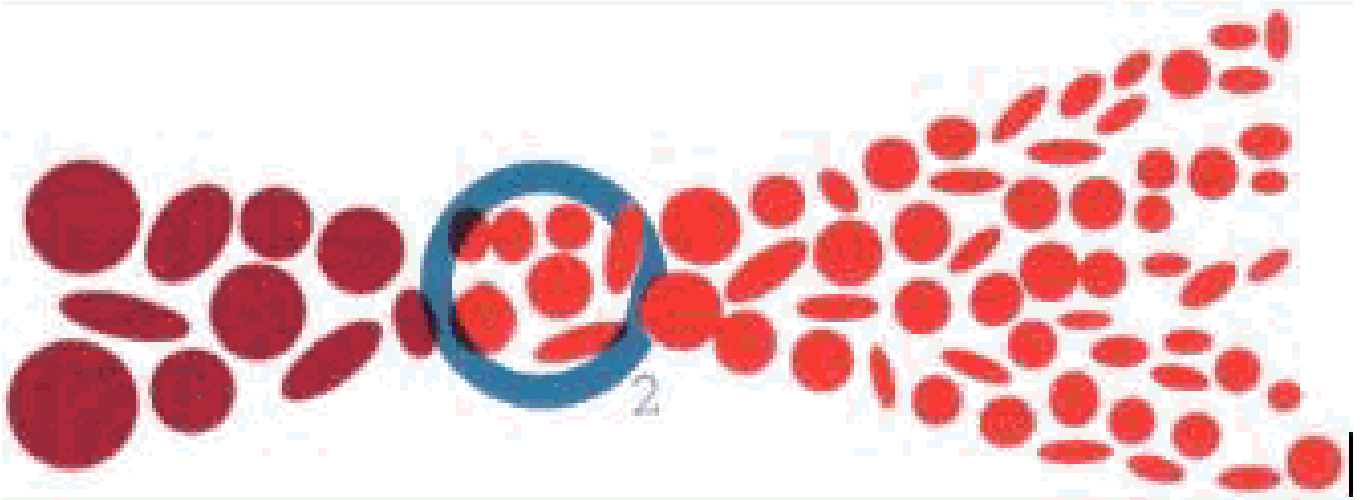
“ Además de la integridad estructural de la pared vascular que forma parte de la microcirculación, el organismo, para mantener su función, necesita de tejido hemático sano inmerso en un sistema coloidal acorde y compatible con la función hidráulica del aparato cardiovascular”

Efecto hemorreológico de la oxivenación

a. Reducción del hematocrito: Volumen de sangre que corresponde a glóbulos rojos.

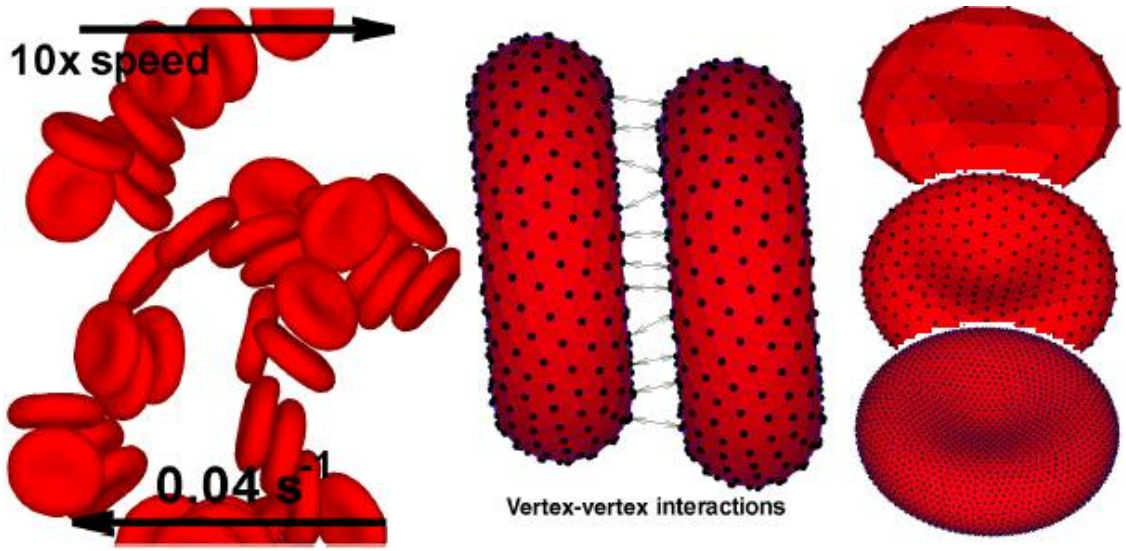
Depende del tamaño y del volumen de glóbulos rojos.





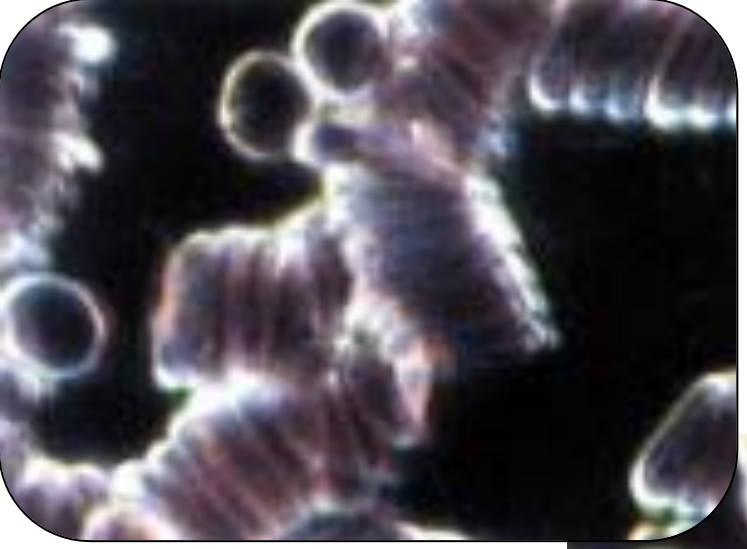
VN: 40 a 45%.

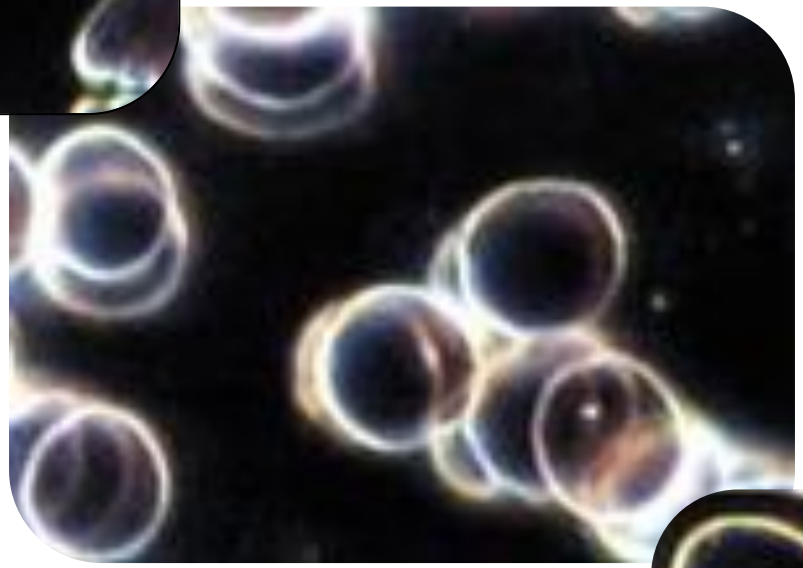
b. Disminución de la fricción endotelial y la resistencia.

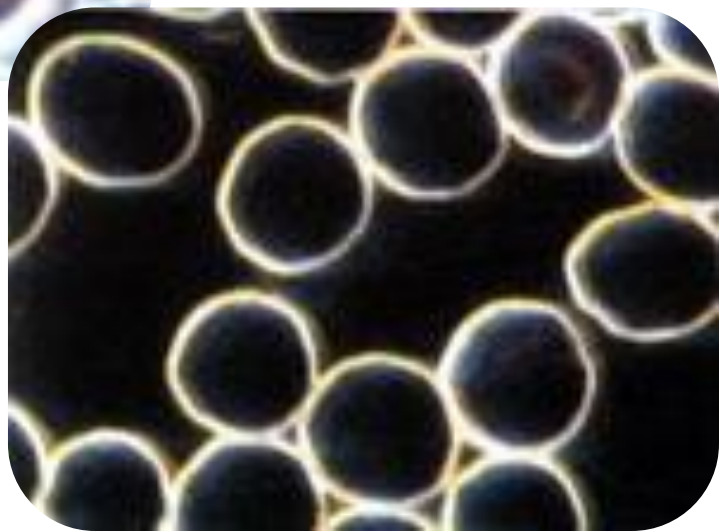


“El endotelio es intocable en condiciones de normalidad. Una capa de flujo laminar de 20 nm de plasma, separa la población celular intravascular circulante, impidiendo el contacto con el endotelio. El estrés por fricción activa genes que inducen engrosamiento difuso de la íntima o ateromas”

Acta médica colombiana Vol 26 No4. Julio-Agosto 2001







http://www.alternativemedizin.de/
oxyvenierung.htm

Pentoxifilina



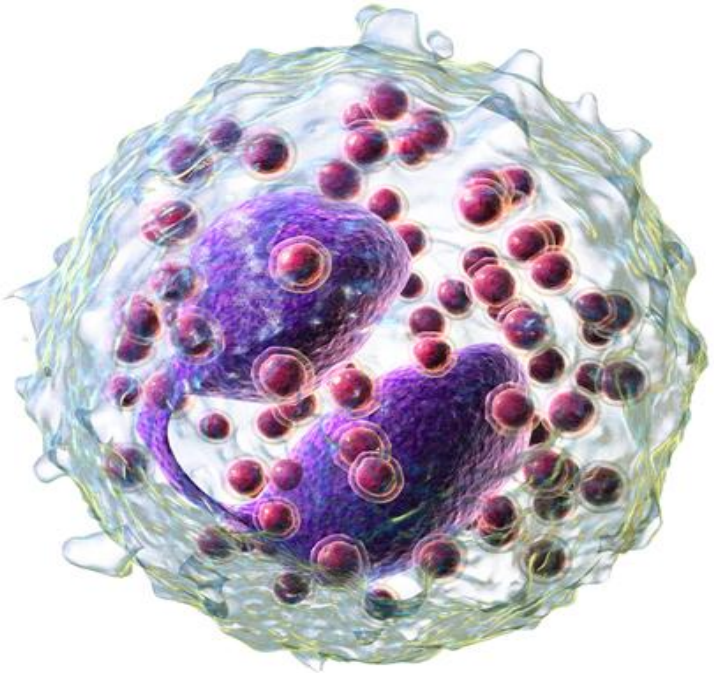
- Famacodinamia:

  Mejora el flujo sanguíneo por incrementar la deformabilidad de los eritrocitos, reduce la agregación de eritrocitos y plaquetas, disminuye los niveles de fibrinógeno, reduce la adherencia de leucocitos al endotelio, reduce la activación de leucocitos y el daño endotelial resultante, y disminuye la viscosidad sanguínea.

  En consecuencia, promueve la microcirculación al mejorar el flujo microcirculatorio y al ejercer efectos antitrombóticos

2. Incremento de Eosinófilos (observado en método Regelberger)

Aumento en el recuento de eosinófilos a 20% después de 20 a 30 sesiones, es decir, 1% por sesión diaria.



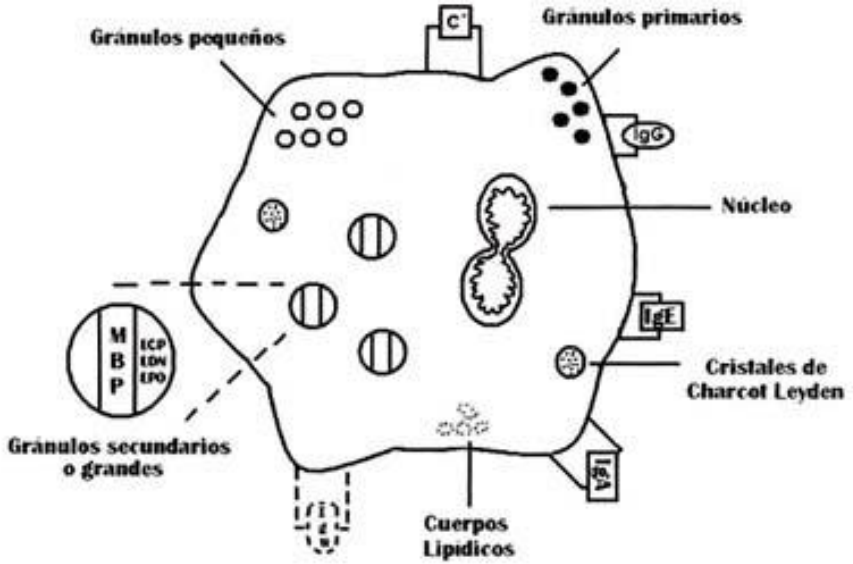
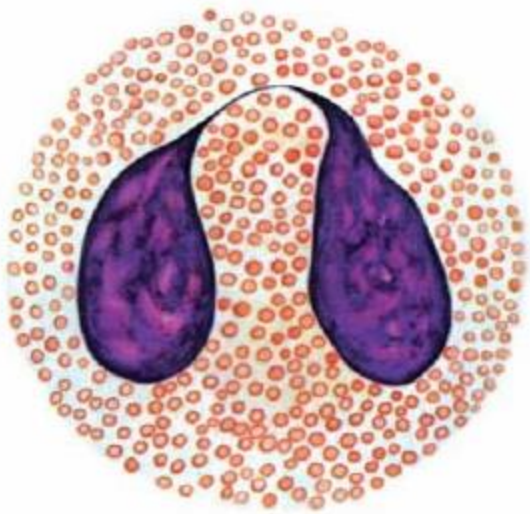


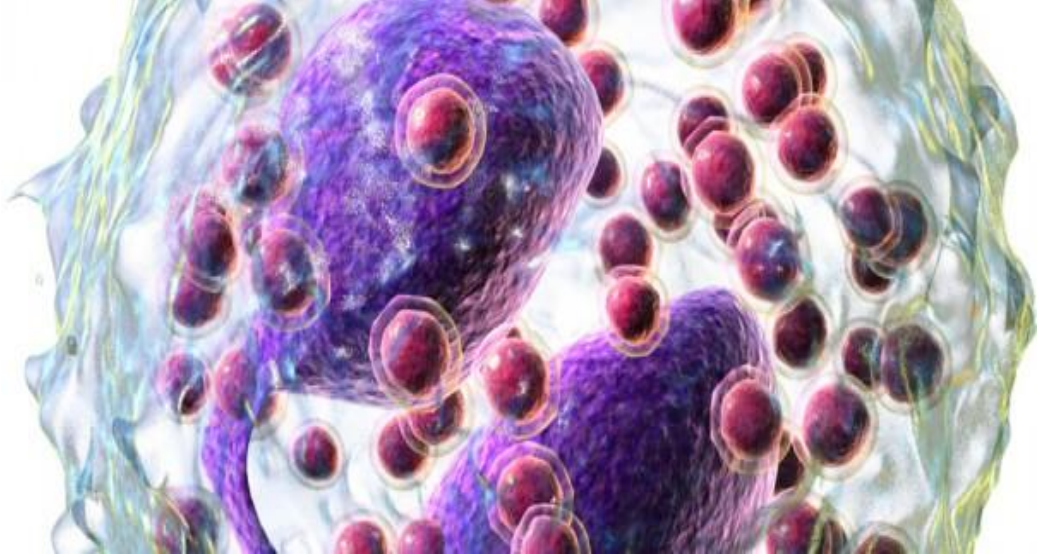
Figura 2: DIAGRAMA DEL EOSINÓFILO: Mostrando sus proteínas granulares y sus eceptores de membrana para las inmunoglobulinas (Ig), Complemento (C')



“Quedó atrás el concepto simplista que de él se tenía: parasitosis y alergias”.

“Actualmente se conoce que desempeñan papel de defensa del huésped frente a microorganismos no fagocitables; función citotóxica inmunorreguladora y participan en la reparación y remodelación tisular (liberando TGF beta)"

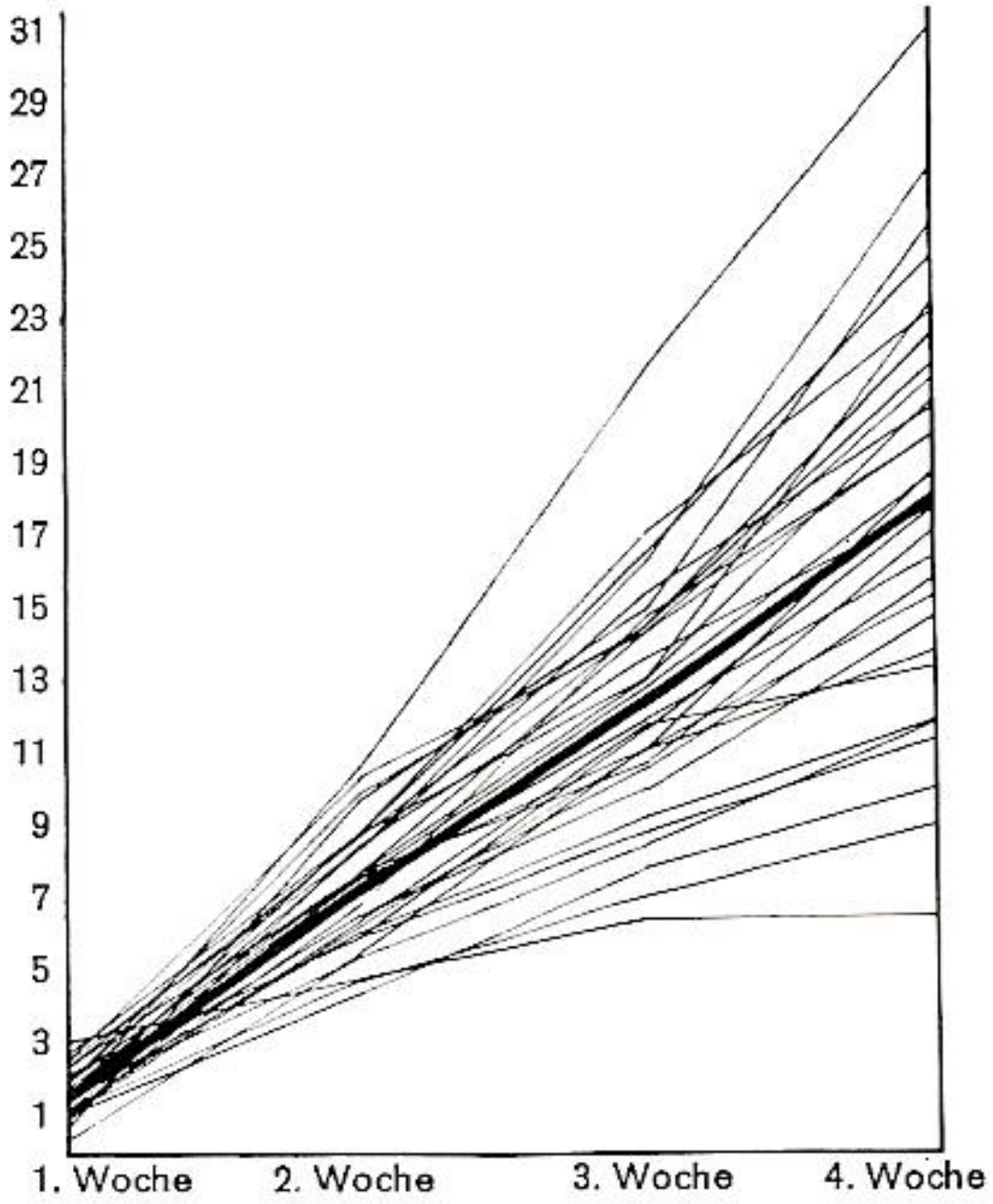
Valdivia L,Dermatología Peruana 2007, Vol 17(2)



- Gránulos primarios: Fosfolipasa A
- Gránulos secundarios o grandes:
  - Peroxidasas: Inducción de degranulación de mastocitos, lisis bacteriana y de helmintos. Acción citotóxica.
  - Proteína Básica Principal: Tóxica para células tumorales y helmintos
  - Proteínas catiónicas: Tóxica para células tumorales y helmintos. Induce necrosis y apoptosis

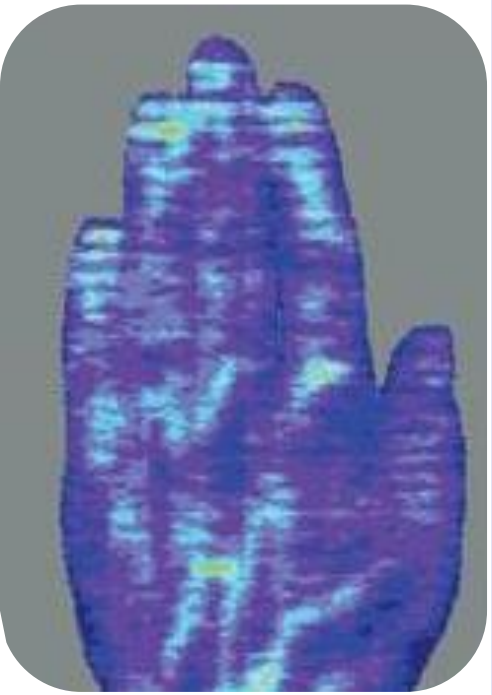
- Gránulos pequeños: Arilsulfatasa B → inactiva leucotrienos

% eosinófilos

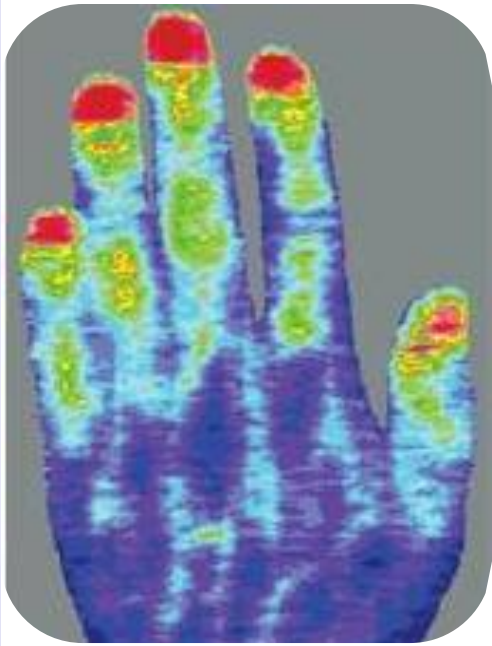


Semanas

3. Aumento de temperatura

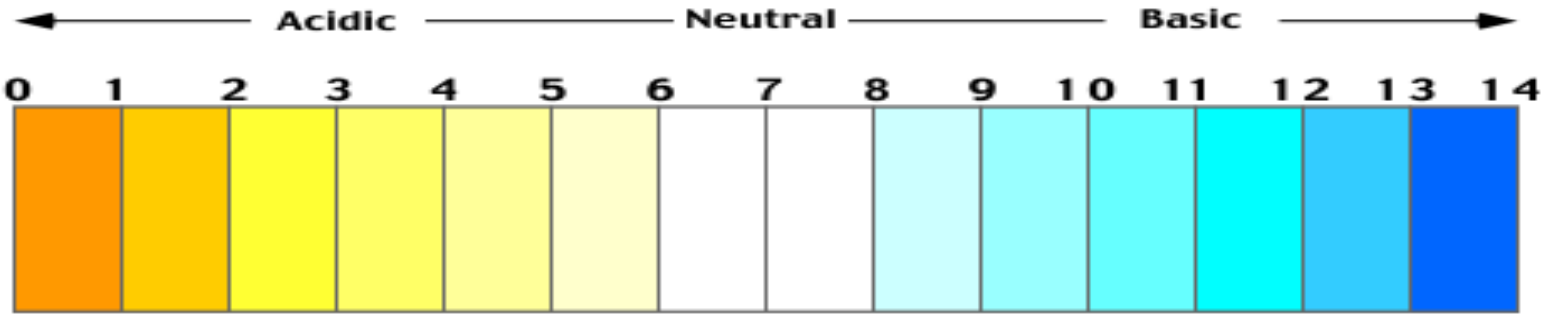


Por termografía se han encontrado diferencias de temperatura entre 0.6 y 0.9 ° C, especialmente en áreas distales. (Aumento de circulación en arteriolas distales, aumento de consumo de oxígeno)



Schwamm, Gengenbach (Thermographie und Herddiagnostik

4. Variación del pH hacia alcalinidad



Hipótesis:

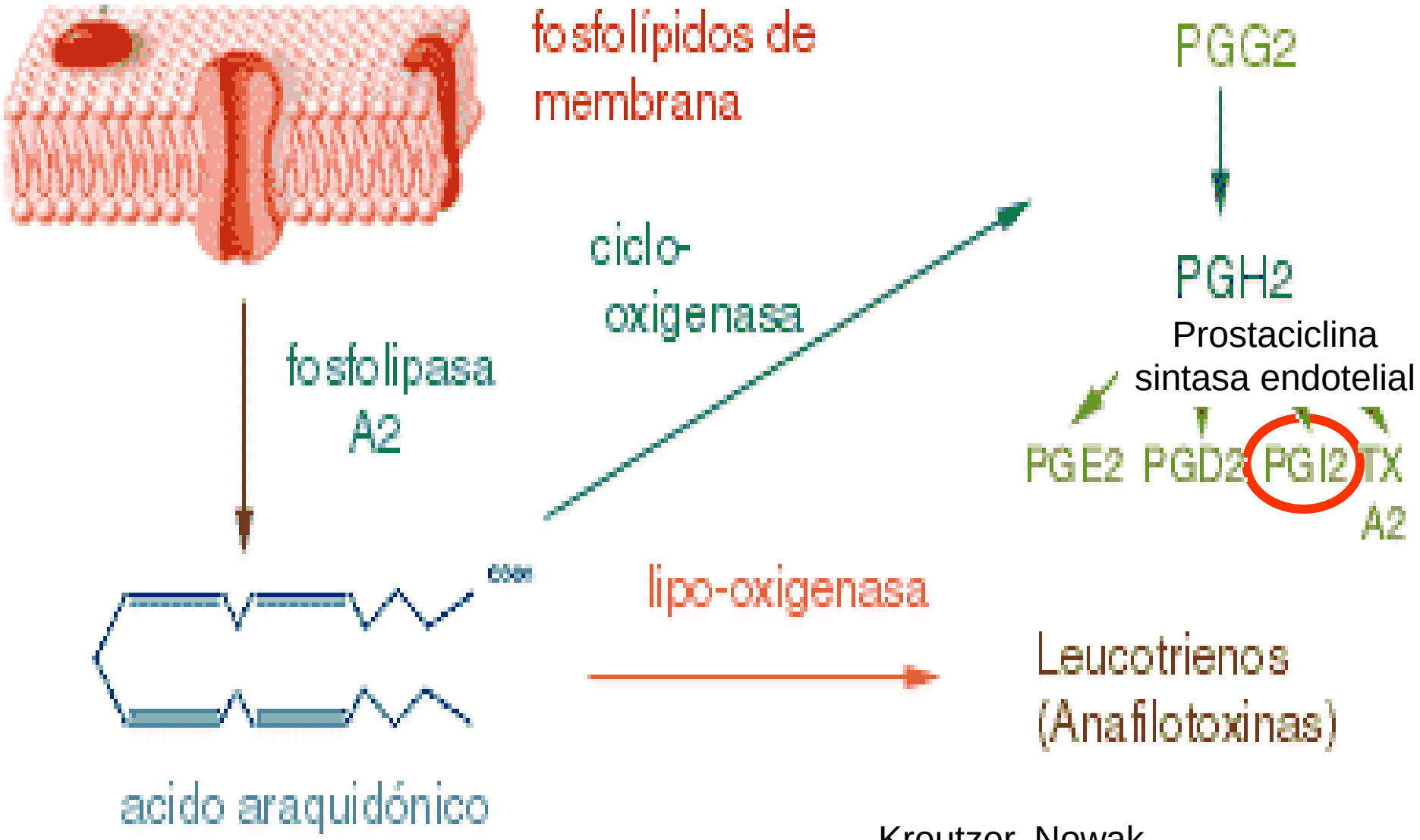
- Aumento de capacidad Buffer de la hemoglobina
- Aumento de la reserva alcalina del plasma

Observable después de 20 aplicaciones

Consecuencias de la Acidosis

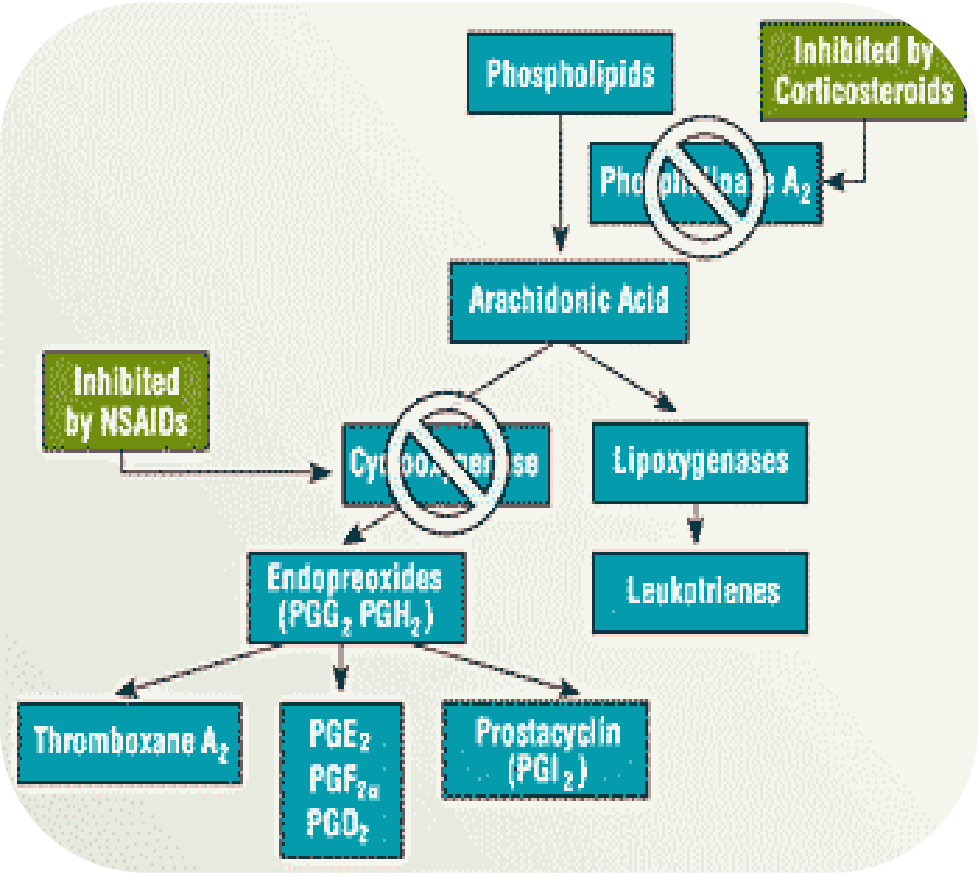
- Disminución de la contractilidad miocárdica.
- Trastornos de la conducción A-V, arritmias.
- Aumento del Ca++ intramiocárdico, en reposo
- Depresión del SNC.
- Aumento de la calcemia por mayor remoción ósea, mayor absorción de fosfatos

5. Aumento de Prostaciclina



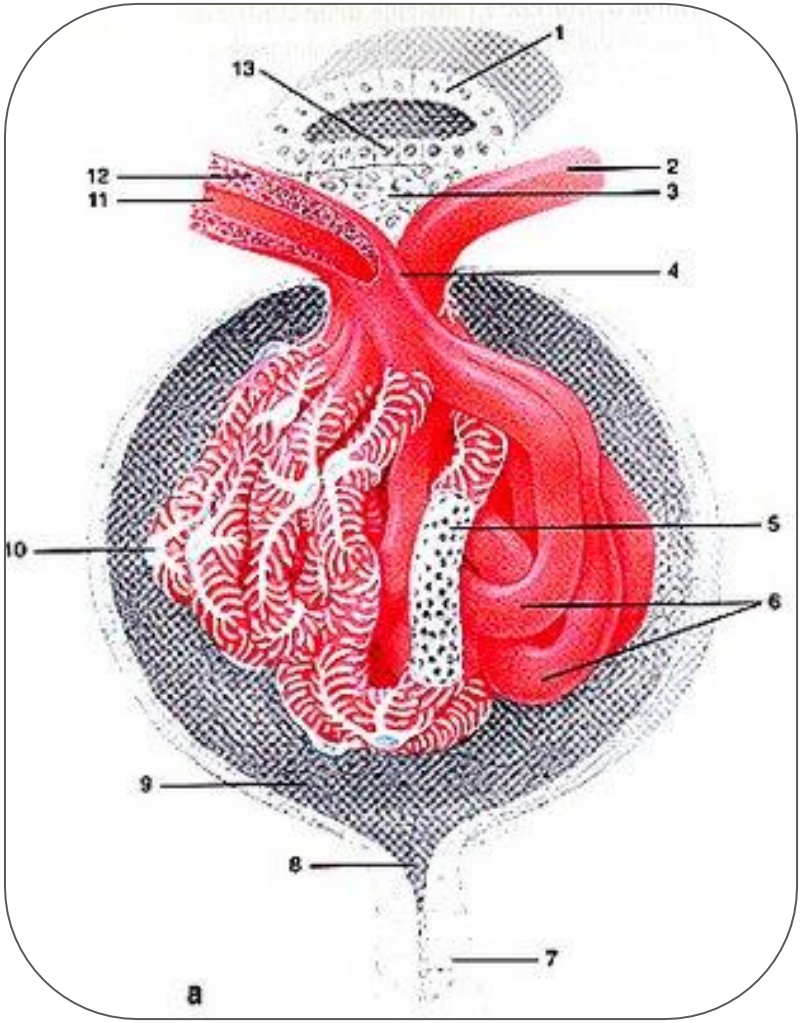
Kreutzer, Nowak

Efectos de la PGI_2



- Vasodilatación por relajación del músculo liso endotelial
- Aumento del flujo sanguíneo
- Antiagregante
- Broncodilatadora
- Vasodilatación pulmonar
- Natriuresis
- Protección endotelial
- Gastroprotección

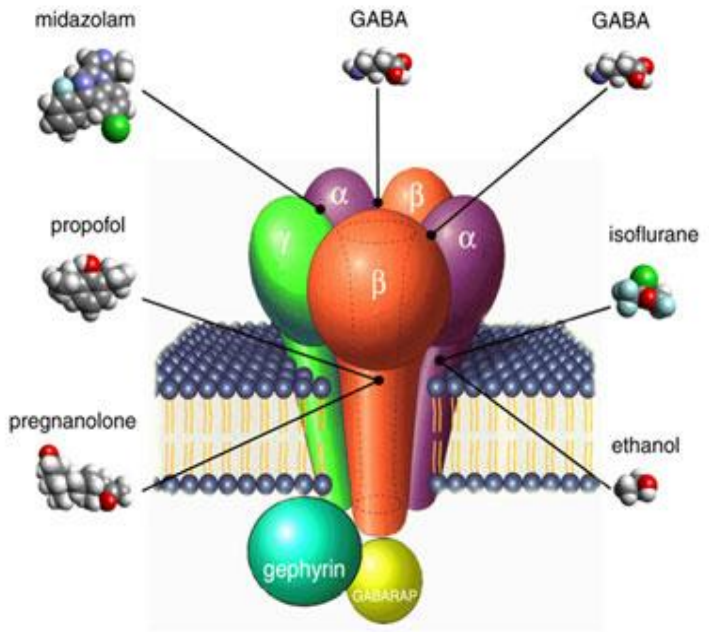
6. Aumento de diuresis



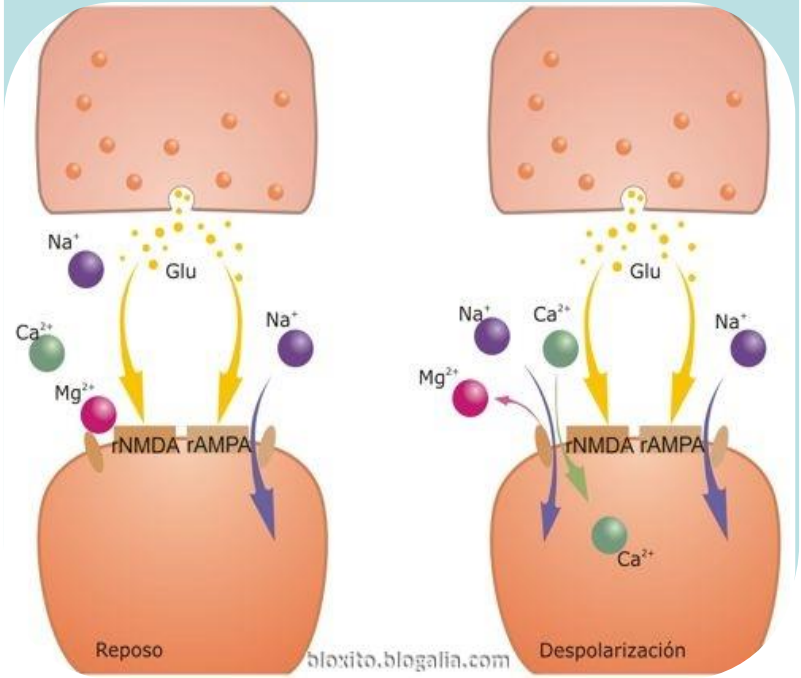
Prostaciclina (PGI_2) y PGE_2 aumentan el flujo sanguíneo y la tasa de filtración glomerular

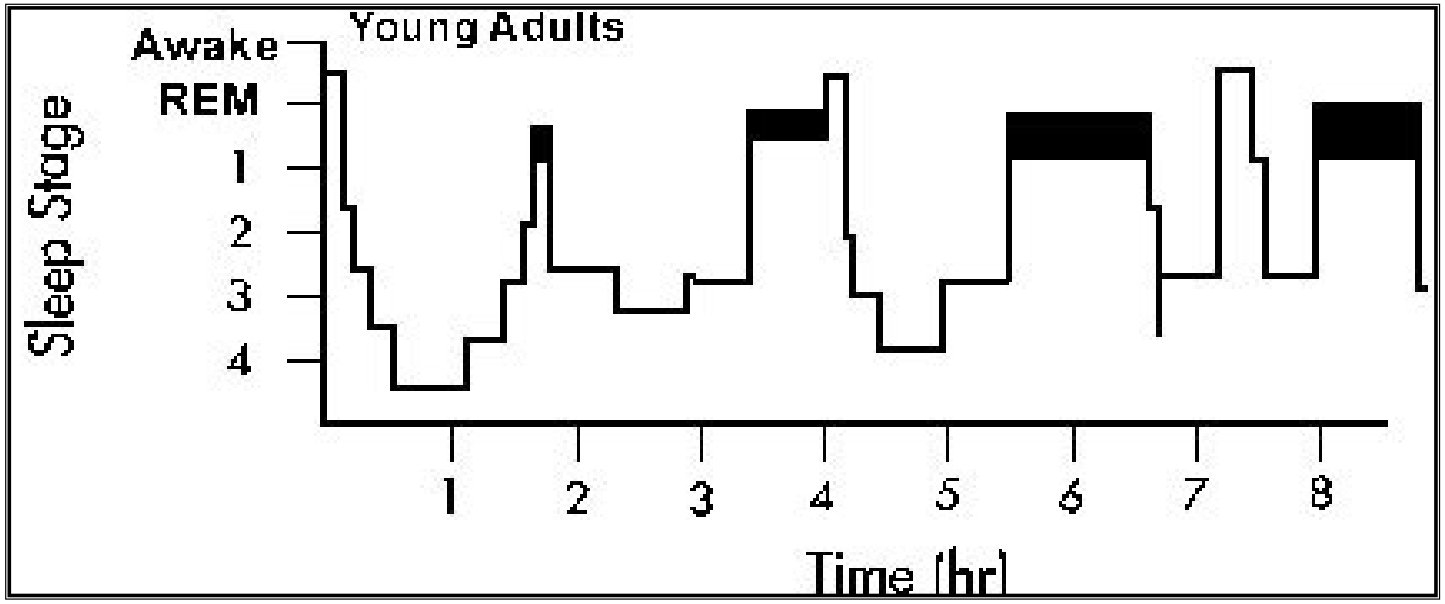
7. Efecto sobre las convulsiones y el sueño

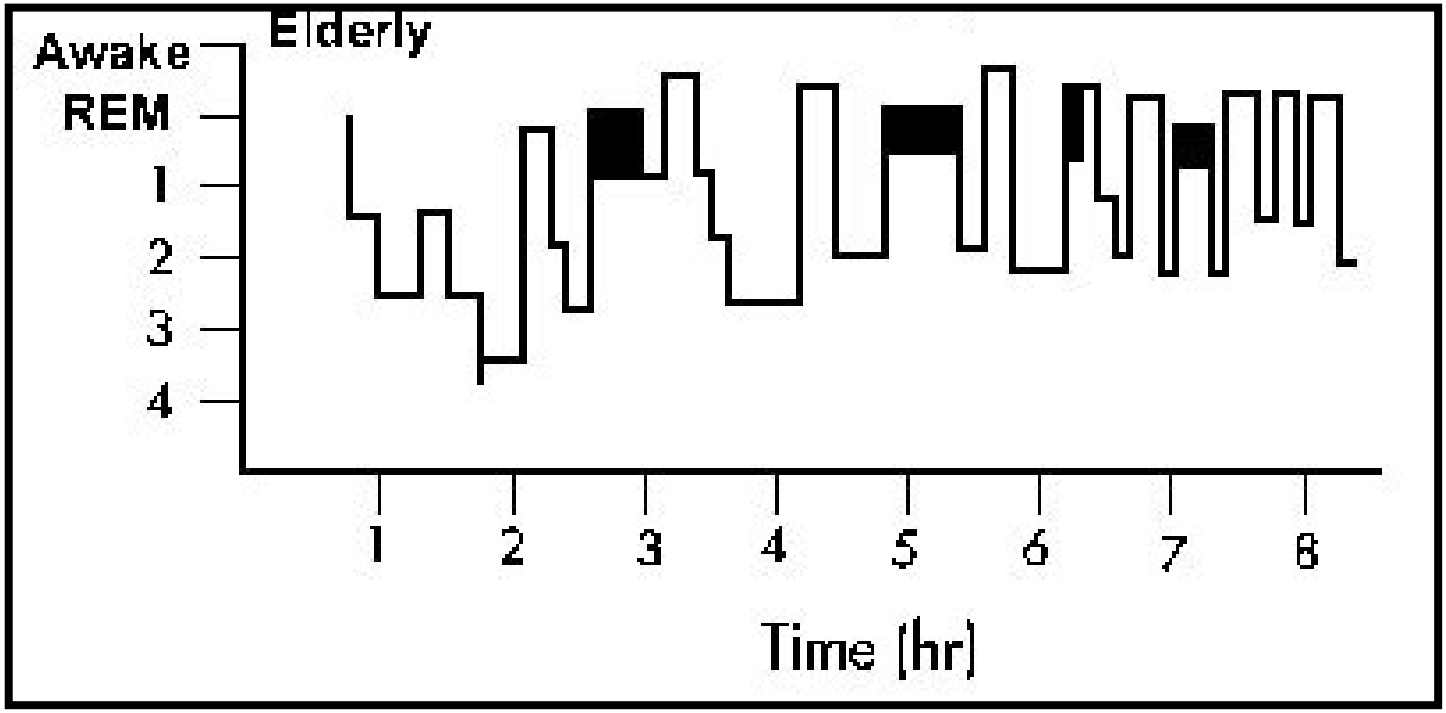
Estímulo de respuestas mediadas por GABA?



Disminución de respuestas mediadas por Glutamato?




Young Adults
Awake
REM
1
2
3
4
Sleep Stage
1
2
3
4
5
6
7
8
Time (hr)


Elderly
Awake
REM
1
2
3
4
1
2
3
4
5
6
7
8
Time (hr)

Oxivenación. Indicaciones
(según Regelsberger)

- **Grupo I:**
  - Alteraciones circulatorias estructurales localizadas en cerebro, ojos, corazón, hígado, riñón, miembros inferiores
  - Alteraciones circulatorias funcionales: migraña y fenómeno de Raynaud.



Oxivenación. Indicaciones
(según Regelsberger)

- **Grupo II:** Enfermedades alérgicas e inflamatorias crónicas. Asma bronquial, bronquitis crónica, dermatitis, hepatitis, inflamaciones de intestino y riñón, artritis.



Indicaciones (Corpas)

- Enfermedad pulmonar que cursa con hipoxemia (con o sin hipercapnia)
- Manifestaciones por hipoxia general o hipoxia localizada.
  - Ej: Trastorno cognitivo del adulto mayor
- Isquemia crónica
  - Sistema nervioso
  - Corazón
  - Miembros inferiores

Indicaciones (Corpas)

- EPOC
- Alteración de la membrana alvéolo capilar
- Hipertensión pulmonar primaria o secundaria
- Síndromes convulsivos
- Cefalea de origen vascular (Como profiláctico; no en dolor agudo)
- Trastornos del sueño (Insomnio)
- Terapia complementaria en SAHOS

- Síndrome vertiginoso
- Enfermedad de Méniere
- Insuficiencia renal crónica
- Hipertensión arterial
- Enfermedades por déficit de ATP
  - Fibromialgia
  - Síndrome de fatiga crónica

Contraindicaciones Método Corpas



- Hemorragia aguda “anormal” (Ejemplo: Hemorragia cerebral)
- Estado febril
- Isquemia aguda (en IAM esperar 3 meses)
- Postoperatorio de cirugía mayor
- Anticoagulantes

- Infección aguda (Ejemplo: Meningitis, encefalitis). Contraindicación en observación

Contraindicaciones adicionales método Regelsberger



- Malformación congénita del corazón (shunt derecha – izquierda)
- Tromboembolismo pulmonar
- Infarto agudo del miocardio (tres meses) (¿riesgo de lesión por reperfusión?)

Precauciones

- Tratamiento con anticoagulantes
- Coagulopatías

Efectos secundarios

- Cefalea
- Enrojecimiento facial
- Tos
- Fatiga
- Disnea y opresión torácica
- Síntomas de resfriado

Frecuencia: 1 a 11%. Con el método Regelsberger

Tratamiento de efectos secundarios

Suspender el goteo

Mantener posición en decúbito

Administrar AINE

Administrar antiinflamatorio esteroideo